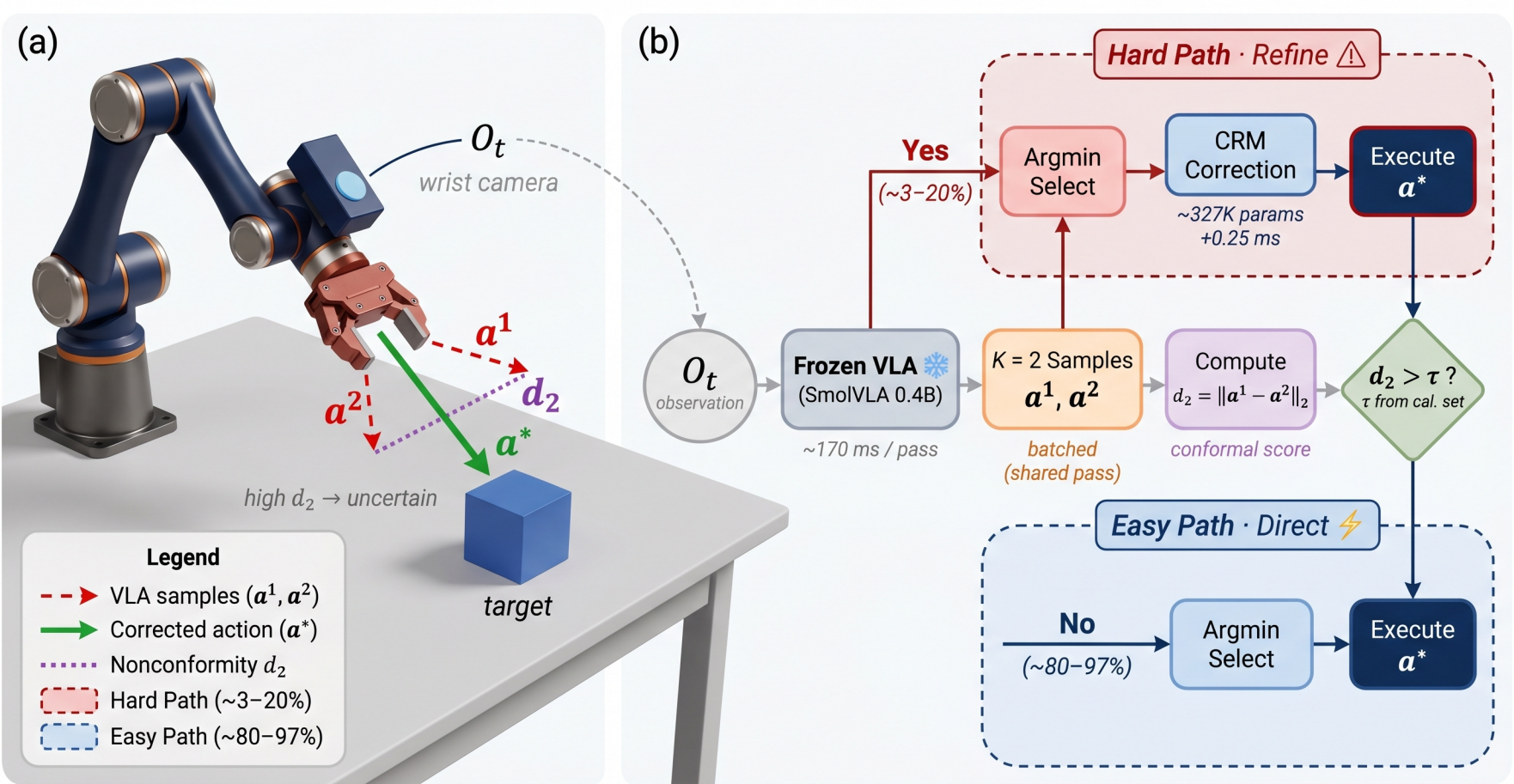

(a)
O_t
wrist camera
a^1
a^2
d_2
a^*
high d_2 → uncertain
target
Legend
VLA samples (a^1, a^2)
Corrected action (a^*)
Nonconformity d_2
Hard Path (~3–20%)
Easy Path (~80–97%)
(b)
Hard Path · Refine ⚠
Yes
(~3–20%)
Argmin Select
CRM Correction
~327K params
+0.25 ms
Execute a^*
O_t
observation
Frozen VLA ❄
(SmolVLA 0.4B)
~170 ms / pass
K = 2 Samples
a^1, a^2
batched
(shared pass)
Compute
$d_2 = \|a^1 - a^2\|_2$
conformal score
$d_2 > \tau$?
τ from cal. set
Easy Path · Direct ⚡
No
(~80–97%)
Argmin Select
Execute a^*